

Titanic Passenger Survival: Visual Analysis and Reporting

- **Course:** ANA 500 Python for Data Science
 - **Assignment:** Micro-Project 2
 - **Name:** Travis James
 - **Dataset:** Titanic passenger data (cleaned in Micro-Project 1)
 - **Scope:** Steps 1 through 4 of the data science process (Acquire, Prepare, Analyze, Report) using Matplotlib and Seaborn
 - **Micro-Project 1:** <https://github.com/OrangeJoe0421/ANA-500-MP1>
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Problem Statement

Survival on the Titanic was not random. This project describes which passenger characteristics (sex, ticket class, age, family situation, fare, and port of embarkation) were associated with who lived and who died.

Hypothesis

Women and passengers in higher ticket classes survived at higher rates than men and passengers in lower ticket classes.

Micro-Project 1 prepared the data. This notebook tests the hypothesis visually, backs each main pattern with a simple statistical check, and reports what the evidence shows. To keep the analysis focused, the hypothesis is broken into three testable parts:

ID	Sub-hypothesis	Main visual
H1	Women survived at a higher rate than men	Survival rate by sex
H2	Survival rate falls from first to third class	Survival rate by class
H3	The sex gap holds inside every class, so class and sex both matter	Sex by class heatmap

Age, family size, fare, and port are treated as supporting variables that may explain or complicate the main result.

Step 1: Acquire

The main input is `titanic_clean.csv`, the output of Micro-Project 1. The raw `titanic.csv` file is also loaded, because Step 2 revisits one cleaning decision (the two missing `Embarked` values) using the original records.

```
In [6]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import matplotlib.ticker as mtick
import seaborn as sns
from scipy.stats import chi2_contingency
from pathlib import Path

pd.set_option("display.max_columns", None)
pd.set_option("display.width", 80)
```

```
In [7]: # Load the cleaned file from Micro-Project 1 and the original raw file.
clean = pd.read_csv("titanic_clean.csv")
raw = pd.read_csv("titanic.csv")

print(f"Cleaned data: {clean.shape[0]:,} rows, {clean.shape[1]} columns")
print(f"Raw data      : {raw.shape[0]:,} rows, {raw.shape[1]} columns")

# CSV files do not store data types, so the category and nullable integer
# types set in Micro-Project 1 need to be restored after loading.
clean["Survived"] = clean["Survived"].astype("Int64")
for col in ["Sex", "Embarked", "Title", "Deck", "AgeGroup"]:
    clean[col] = clean[col].astype("category")

clean.head()
```

```
Cleaned data: 1,309 rows, 20 columns
Raw data      : 1,309 rows, 12 columns
```

Out[7]:

	PassengerId	Survived	Pclass	Sex	Age	SibSp	Parch	Fare	Embarked	Name
0	1	0	3	male	22.0	1	0	7.2500	S	Braund, Mr. Owen Harris
1	2	1	1	female	38.0	1	0	71.2833	C	Cumings, Mrs. John Bradley (Florence Briggs Th.)
2	3	1	3	female	26.0	0	0	7.9250	S	Heikkinen, Miss. Laina
3	4	1	1	female	35.0	1	0	53.1000	S	Futrelle, Mrs. Jacques Heath (Lily May Peel)
4	5	0	3	male	35.0	0	0	8.0500	S	Allen, Mr. William Henry

In [8]:

```
# Quick integrity check that the file loaded matches what Micro-Project 1
# saved: same row count, unique IDs, and only Survived has blanks.
assert len(clean) == 1309, "Row count does not match Micro-Project 1"
assert clean["PassengerId"].is_unique, "PassengerId is not unique"
missing = clean.isna().sum()
assert set(missing[missing > 0].index) == {"Survived"}, "Unexpected blanks"
print("Cleaned data matches the Micro-Project 1 output.")
```

Cleaned data matches the Micro-Project 1 output.

Step 2: Prepare

Most preparation was completed in Micro-Project 1. This step does three things before analysis:

1. Revisits the **Embarked** imputation based on instructor feedback
2. Builds the analysis set of passengers with a known outcome
3. Adds readable labels and sets a consistent chart style

2a. Revisiting the two missing **Embarked** values

In Micro-Project 1 the two missing ports were filled with the overall mode (S, Southampton). The feedback asked whether the shared ticket and cabin could support a more specific imputation. The checks below test that.

```
In [9]: # Pull the two passengers with no port from the raw file.
no_port = raw.loc[raw["Embarked"].isna(),
                  ["PassengerId", "Name", "Pclass", "Ticket", "Fare", "Cabin"]]
print(no_port.to_string(index=False))
```

```
ticket = no_port["Ticket"].iloc[0]
cabin = no_port["Cabin"].iloc[0]
print(f"\nPassengers on ticket {ticket}:", (raw["Ticket"] == ticket).sum())
print(f"Passengers in cabin {cabin} :", (raw["Cabin"] == cabin).sum())
```

PassengerId	Name	Pclass	Ticket	Fare	Cabin
62	Icard, Miss. Amelie	1	113572	80.0	B28
830	Stone, Mrs. George Nelson (Martha Evelyn)	1	113572	80.0	B28

Passengers on ticket 113572: 2

Passengers in cabin B28 : 2

Both passengers share ticket 113572 and cabin B28, but no one else in the file does, so there is no travel companion whose port can simply be copied. The next best evidence is passengers who look like them: first class, a similar ticket number, and a similar fare.

```
In [10]: series_113 = raw["Ticket"].str.match(r"^113\d{3}$")
print("Ports for other 113xxx tickets:")
print(raw.loc[series_113 & raw["Embarked"].notna(), "Embarked"]
       .value_counts())

raw["TicketGroup"] = raw.groupby("Ticket")["Ticket"].transform("size")
raw["FarePerPerson"] = raw["Fare"] / raw["TicketGroup"]
first = raw[raw["Pclass"] == 1]
print("\nMedian first-class fare per person by port:")
print(first.groupby("Embarked")["FarePerPerson"].median().round(2))
similar = first["FarePerPerson"].between(35, 45)
print("\nPorts for first-class passengers paying 35 to 45 per person:")
print(first.loc[similar, "Embarked"].value_counts())
```

Ports for other 113xxx tickets:

Embarked

S 54

C 10

Name: count, dtype: int64

Median first-class fare per person by port:

Embarked

C 34.65

Q 30.00

S 26.55

Name: FarePerPerson, dtype: float64

Ports for first-class passengers paying 35 to 45 per person:

Embarked

C 45

S 27

Name: count, dtype: int64

Decision: the two lines of evidence point in different directions. Fare per person leans toward Cherbourg (C), but only loosely, because first-class fares from C and S overlap heavily. The ticket series is the stronger signal: about 84% of other 113xxx tickets (54 of 64) boarded at Southampton, and a ticket number is tied to how and where the ticket was bought, while a fare can be matched by passengers from any port. The two passengers are therefore assigned **S** based on their ticket series.

The final value is the same as in Micro-Project 1, but it is now justified by evidence specific to these passengers rather than the overall mode. The imputation is recorded in a new flag column so it stays traceable.

```
In [11]: # Apply the ticket-series imputation and flag the two rows.
fill_ids = no_port["PassengerId"]
clean["EmbarkedWasMissing"] = clean["PassengerId"].isin(fill_ids)
clean.loc[clean["EmbarkedWasMissing"], "Embarked"] = "S"
assert (clean.loc[clean["EmbarkedWasMissing"], "Embarked"] == "S").all()
print("Rows flagged as imputed Embarked:", clean["EmbarkedWasMissing"].sum())
```

Rows flagged as imputed Embarked: 2

2b. Build the analysis set

```
In [12]: # Only the 891 passengers with a known outcome can test the hypothesis. The
# 418 holdout rows (blank Survived) are set aside, as planned in
# Micro-Project 1.
df = clean[clean["Survived"].notna()].copy()
df["Survived"] = df["Survived"].astype(int)
print(f"Analysis set: {len(df)} passengers")
print(f"Overall survival rate: {df['Survived'].mean():.1%}")

# Readable labels for chart axes and legends. The numeric and coded columns
# stay unchanged for calculations.
df["Outcome"] = df["Survived"].map({0: "Died", 1: "Survived"})
df["Class"] = df["Pclass"].map({1: "1st", 2: "2nd", 3: "3rd"})
```

```
df["Port"] = df["Embarked"].map(
    {"C": "Cherbourg", "Q": "Queenstown", "S": "Southampton"})
df["SexLabel"] = df["Sex"].astype(str).str.title()

# Fixed orders so every chart lists categories the same way.
CLASS_ORDER = ["1st", "2nd", "3rd"]
SEX_ORDER = ["Female", "Male"]
AGE_ORDER = ["Child", "Teen", "Adult", "MiddleAge", "Senior"]
PORT_ORDER = ["Cherbourg", "Queenstown", "Southampton"]
```

Analysis set: 891 passengers

Overall survival rate: 38.4%

2c. Chart style

```
In [13]: BLUE, ORANGE, AQUA = "#2a78d6", "#eb6834", "#1baf7a"
INK, MUTED, GRID = "#0b0b0b", "#52514e", "#e4e3df"
OUTCOME_COLORS = {"Survived": BLUE, "Died": ORANGE}
SEX_COLORS = {"Female": BLUE, "Male": ORANGE}

sns.set_theme(style="whitegrid")
plt.rcParams.update({
    "figure.dpi": 110, "savefig.dpi": 200, "savefig.bbox": "tight",
    "axes.edgecolor": MUTED, "axes.labelcolor": INK, "text.color": INK,
    "xtick.color": MUTED, "ytick.color": MUTED, "grid.color": GRID,
    "axes.spines.top": False, "axes.spines.right": False,
    "axes.grid.axis": "y", "axes.titlesize": 13,
    "axes.titleweight": "bold", "axes.labelsize": 11,
})

# Every chart is also saved as a PNG for the slide deck.
FIG_DIR = Path("figures")
FIG_DIR.mkdir(exist_ok=True)

def labelBars(ax, fmt="{:.0%}", counts=None):
    for i, bar in enumerate(ax.patches):
        h = bar.get_height()
        if np.isnan(h):
            continue
        text = fmt.format(h)
        if counts is not None:
            text += f"\nn={counts[i]}"
        ax.annotate(text, (bar.get_x() + bar.get_width() / 2, h),
                    ha="center", va="bottom", fontsize=9, color=INK,
                    xytext=(0, 3), textcoords="offset points")

def pct_axis(ax, top=1.0):
    # Format the y-axis as percentages with a little headroom for labels.
    ax.yaxis.set_major_formatter(mtick.PercentFormatter(1.0))
    ax.set_ylim(0, top)
```

Step 3: Analyze

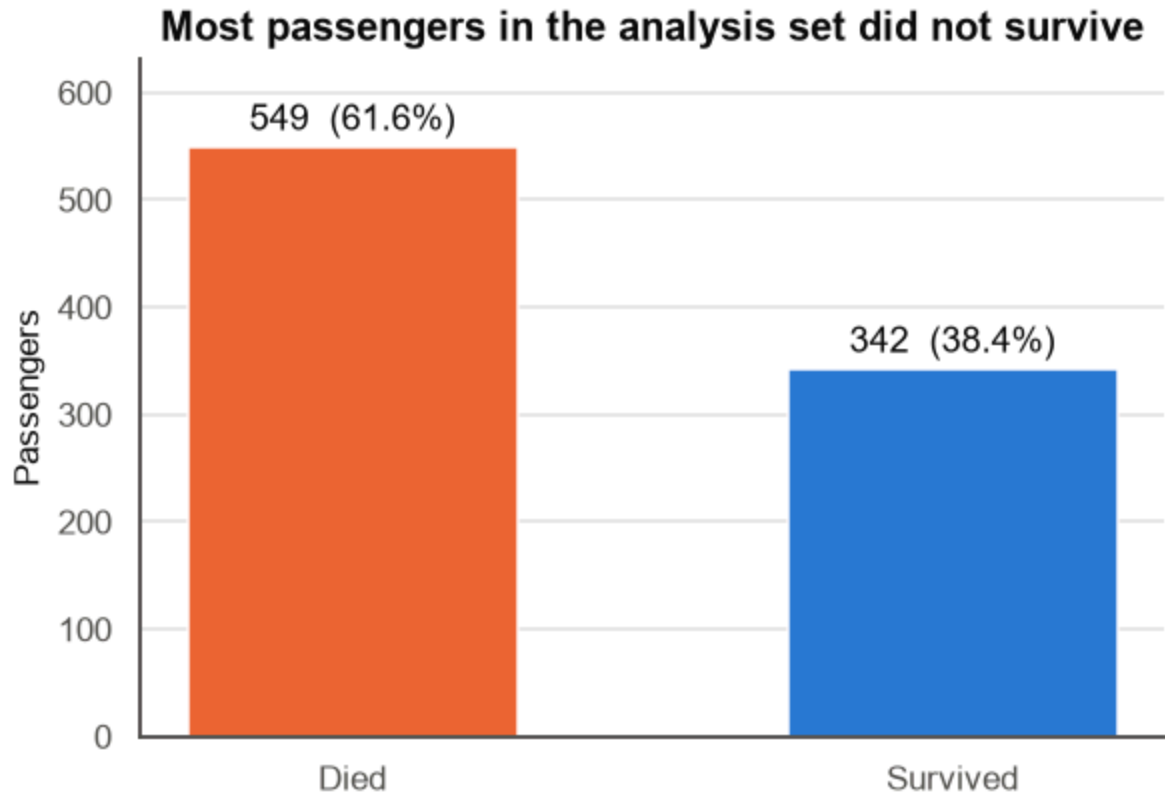
Analytical techniques used

- **Grouped survival rates** (bar charts): the share of each group that survived, compared against the overall rate of 38.4%
- **Two-way comparison** (heatmap): survival rate for every sex and class combination, to check whether one factor explains the other
- **Distributions** (histograms and box plots): how age and fare differ between passengers who lived and died
- **Chi-square test of independence**: a statistical check that each visual gap is too large to be explained by chance
- **Correlation matrix**: a compact view of which numeric features move with survival

3a. Baseline: overall survival

```
In [14]: # Count how many passengers died and survived, and show it as a bar chart.
BASE = df["Survived"].mean()
outcome_counts = df["Outcome"].value_counts().reindex(["Died", "Survived"])

fig, ax = plt.subplots(figsize=(6, 4))
ax.bar(outcome_counts.index, outcome_counts.values,
       color=[ORANGE, BLUE], width=0.55)
for i, v in enumerate(outcome_counts.values):
    ax.annotate(f"{v} ({v / len(df):.1%})", (i, v), ha="center",
               va="bottom", xytext=(0, 3), textcoords="offset points")
ax.set_title("Most passengers in the analysis set did not survive")
ax.set_ylabel("Passengers")
ax.set_ylim(0, outcome_counts.max() * 1.15)
fig.savefig(FIG_DIR / "01_overall.png")
plt.show()
```



Observation: 549 of 891 passengers (61.6%) died and 342 (38.4%) survived. Any group whose survival rate is well above or below 38.4% stands out from the baseline.

3b. H1 and H2: survival by sex and by ticket class

```
In [15]: sex_rate = df.groupby("SexLabel")["Survived"].agg(["mean", "size"])
sex_rate = sex_rate.reindex(SEX_ORDER)
class_rate = df.groupby("Class")["Survived"].agg(["mean", "size"])
class_rate = class_rate.reindex(CLASS_ORDER)

fig, axes = plt.subplots(1, 2, figsize=(11, 4.5), sharey=True)

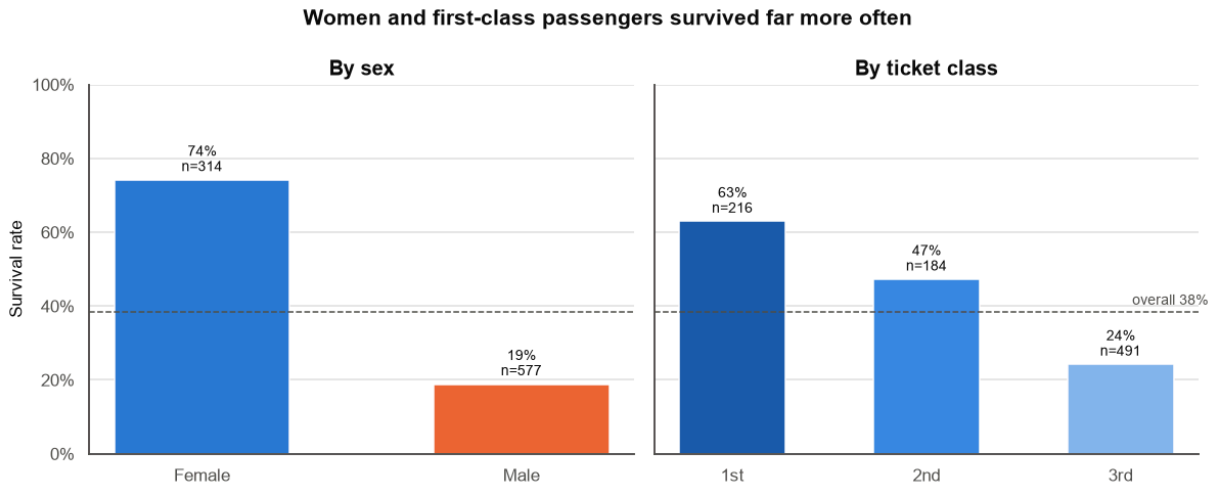
axes[0].bar(sex_rate.index, sex_rate["mean"],
            color=[SEX_COLORS[s] for s in sex_rate.index], width=0.55)
label_bars(axes[0], counts=sex_rate["size"].tolist())
axes[0].set_title("By sex")

axes[1].bar(class_rate.index, class_rate["mean"],
            color=["#1c5cab", "#3987e5", "#86b6ef"], width=0.55)
label_bars(axes[1], counts=class_rate["size"].tolist())
axes[1].set_title("By ticket class")

for ax in axes:
    ax.axhline(BASE, ls="--", lw=1, color=MUTED)
    pct_axis(ax, 1.0)
axes[0].set_ylabel("Survival rate")
axes[1].text(2.45, BASE + 0.02, "overall 38%", ha="right",
            fontsize=9, color=MUTED)
```

```
fig.suptitle("Women and first-class passengers survived far more often",
            fontsize=14, fontweight="bold")
fig.tight_layout()
fig.savefig(FIG_DIR / "02_sex_class.png")
plt.show()

print(sex_rate.assign(mean=sex_rate["mean"].round(3)))
print()
print(class_rate.assign(mean=class_rate["mean"].round(3)))
```



	mean	size
SexLabel		
Female	0.742	314
Male	0.189	577

	mean	size
Class		
1st	0.630	216
2nd	0.473	184
3rd	0.242	491

Observation (H1): 74.2% of women survived compared with 18.9% of men, a gap of about 55 percentage points. Women were only 35% of the analysis set but made up 68% of the survivors (233 of 342).

Observation (H2): survival falls steadily with class: 63.0% in first, 47.3% in second, and 24.2% in third. Third class was the largest group (491 passengers) and the only class below the overall rate.

3c. H3: does the sex gap hold inside every class?

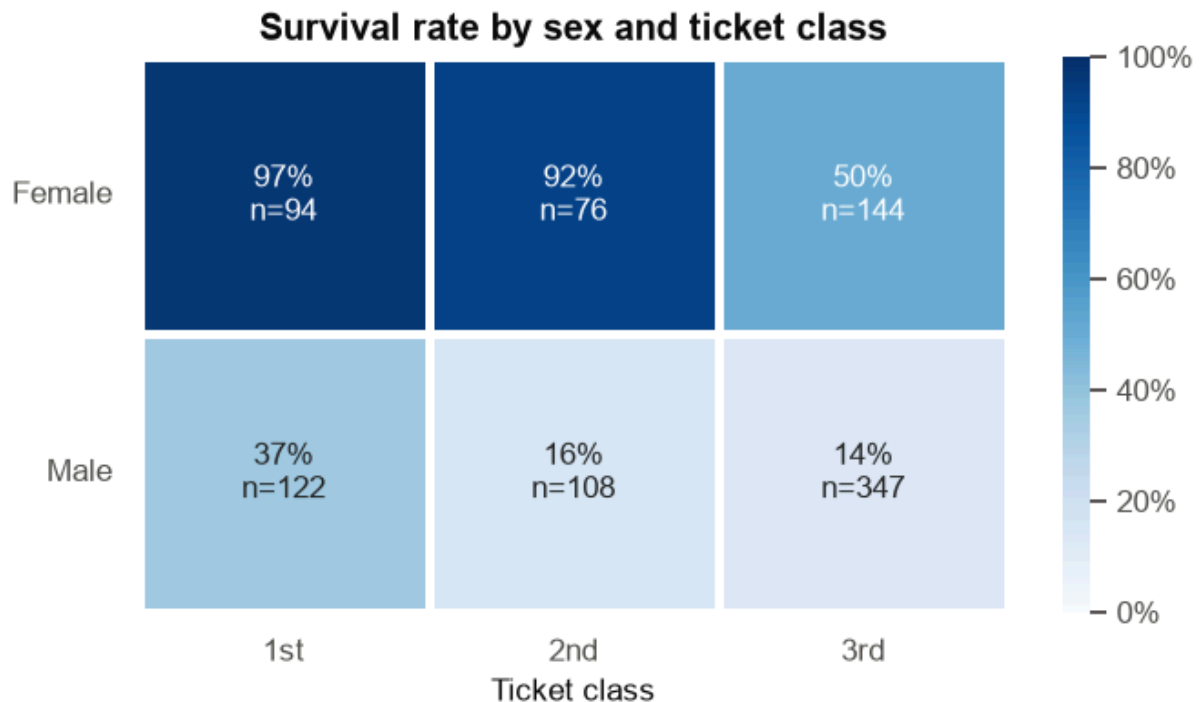
```
In [16]: rate = df.pivot_table(index="SexLabel", columns="Class", values="Survived",
                               aggfunc="mean").reindex(index=SEX_ORDER,
                                                       columns=CLASS_ORDER)
size = df.pivot_table(index="SexLabel", columns="Class", values="Survived",
                      aggfunc="size").reindex(index=SEX_ORDER,
                                              columns=CLASS_ORDER)
annot = rate.map(lambda v: f"{v:.0%}") + "\nn=" + size.astype(str)
```

```

fig, ax = plt.subplots(figsize=(7, 3.6))
sns.heatmap(rate, annot=annot, fmt="", cmap="Blues", vmin=0, vmax=1,
            linewidths=2, linecolor="white",
            cbar_kws={"format": mtick.PercentFormatter(1.0)},
            annot_kws={"fontsize": 11}, ax=ax)
ax.grid(False)
ax.set_title("Survival rate by sex and ticket class")
ax.set_xlabel("Ticket class")
ax.set_ylabel("")
ax.tick_params(axis="y", rotation=0)
fig.savefig(FIG_DIR / "03_heatmap.png")
plt.show()

rate.round(3)

```



Out[16]:

Class	1st	2nd	3rd
SexLabel			
Female	0.968	0.921	0.500
Male	0.369	0.157	0.135

Observation (H3): women out-survived men in every class, and both sexes did better in higher classes. The two factors stack:

- First- and second-class women survived at 97% and 92%.
- Third-class women survived at 50%, well below other women but still above first-class men (37%).
- Second- and third-class men were the worst off at 16% and 14%.

So class alone does not explain the sex gap, and sex alone does not explain the class gap. Sex is the stronger factor, since the lowest female group beats the highest male group.

3d. Statistical check: chi-square tests

```
In [17]: def chi_square(col):
    table = pd.crosstab(df[col], df["Survived"])
    chi2, p, dof, _ = chi2_contingency(table)
    n = table.to_numpy().sum()
    v = np.sqrt(chi2 / (n * (min(table.shape) - 1)))
    return {"variable": col, "chi2": round(chi2, 1), "dof": dof,
            "p_value": p, "cramers_v": round(v, 3)}

tests = pd.DataFrame([chi_square(c) for c in
                      ["Sex", "Pclass", "Title", "Port", "AgeGroup",
                       "IsAlone"]])
tests = tests.sort_values("cramers_v", ascending=False)
tests["p_value"] = tests["p_value"].map(lambda p: f"{p:.1e}")
tests
```

```
Out[17]:
```

	variable	chi2	dof	p_value	cramers_v
2	Title	288.1	4	4.0e-61	0.569
0	Sex	260.7	1	1.2e-58	0.541
1	Pclass	102.9	2	4.5e-23	0.340
5	IsAlone	36.0	1	2.0e-09	0.201
3	Port	26.0	2	2.3e-06	0.171
4	AgeGroup	17.1	4	1.8e-03	0.139

Observation: every tested variable is associated with survival at far below the usual 0.05 cutoff, but the strength differs a lot. Sex ($V = 0.54$) and title are the strongest. Class ($V = 0.34$) is next. Port, age group, and traveling alone are much weaker. This ranking matches the charts and supports the hypothesis that sex and class are the main drivers.

3e. Supporting variable: age

```
In [18]: known_age = df[~df["AgeWasMissing"]]
print(f"Passengers with a recorded age: {len(known_age)}")

fig, axes = plt.subplots(1, 2, figsize=(12, 4.5))

sns.histplot(data=known_age, x="Age", hue="Outcome", bins=16,
             hue_order=["Died", "Survived"], palette=OUTCOME_COLORS,
             multiple="dodge", shrink=0.85, alpha=1, edgecolor="white",
             ax=axes[0])
```

```

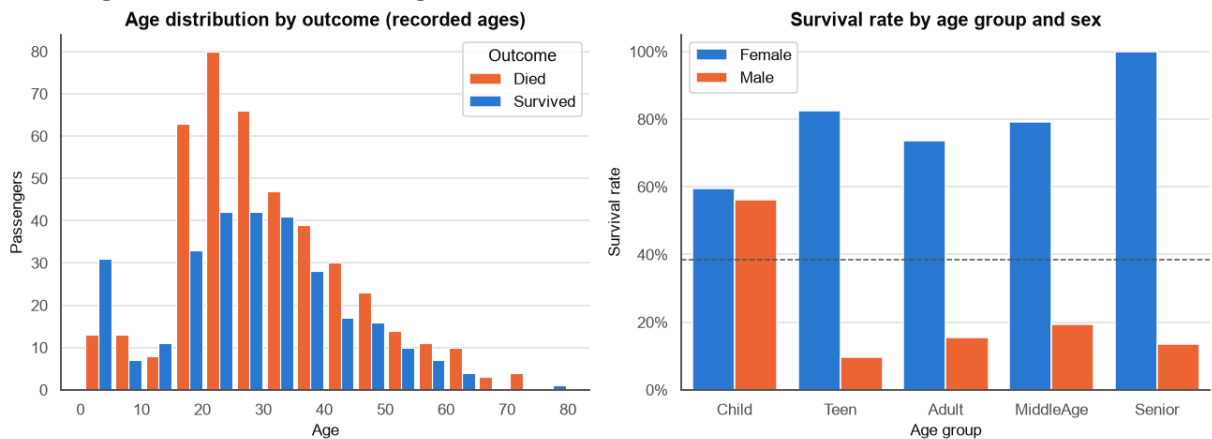
axes[0].set_title("Age distribution by outcome (recorded ages)")
axes[0].set_ylabel("Passengers")

sns.barplot(data=df, x="AgeGroup", y="Survived", hue="SexLabel",
            order=AGE_ORDER, hue_order=SEX_ORDER, palette=SEX_COLORS,
            errorbar=None, saturation=1, ax=axes[1])
axes[1].axhline(BASE, ls="--", lw=1, color=MUTED)
pct_axis(axes[1], 1.05)
axes[1].set_title("Survival rate by age group and sex")
axes[1].set_xlabel("Age group")
axes[1].set_ylabel("Survival rate")
axes[1].legend(title="")
fig.tight_layout()
fig.savefig(FIG_DIR / "04_age.png")
plt.show()

df.pivot_table(index="AgeGroup", columns="SexLabel", values="Survived",
               aggfunc="mean", observed=True).reindex(AGE_ORDER).round(3)

```

Passengers with a recorded age: 714



Out[18]:

SexLabel	Female	Male
AgeGroup		
Child	0.594	0.561
Teen	0.826	0.095
Adult	0.736	0.156
MiddleAge	0.793	0.194
Senior	1.000	0.136

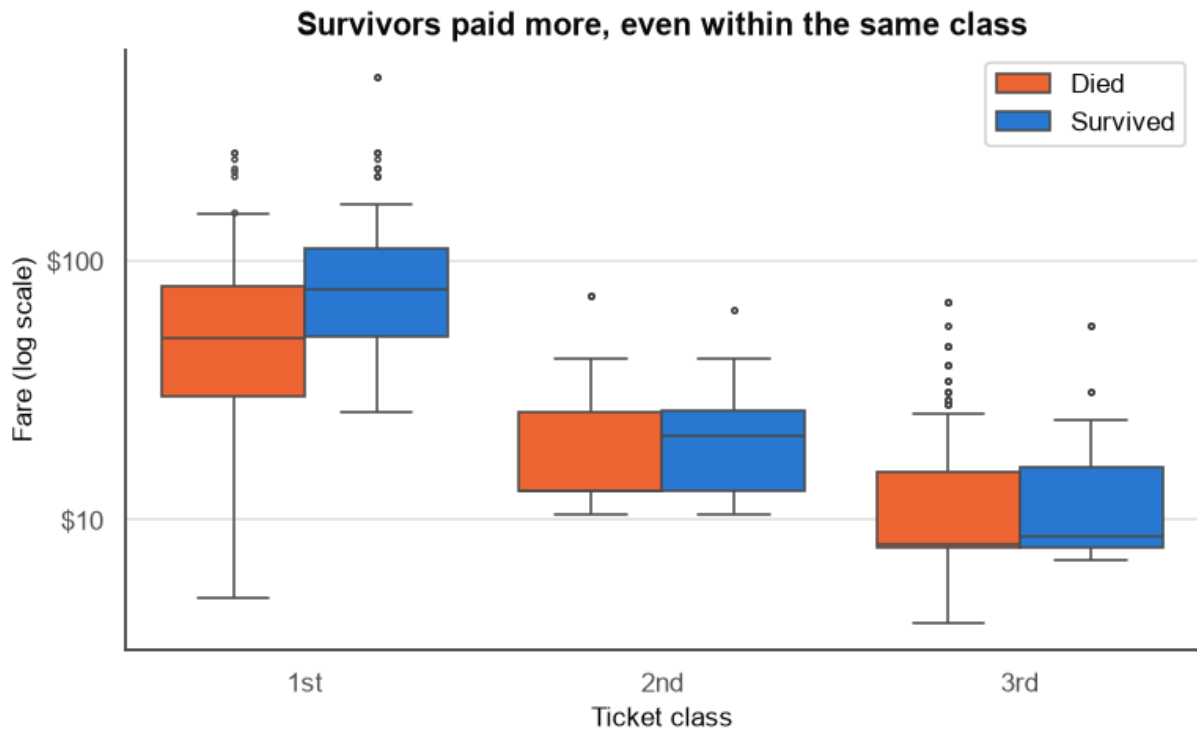
Observation: the histogram shows one clear age effect: young children survived more often than they died, while the 20 to 30 range is dominated by deaths. Split by sex, the picture is sharper. Boys under 13 survived at about 56%, far above adult men (about 16%), which fits a "women and children first" pattern. For women, survival stayed high at every age. Age matters, but mainly for males, and the effect is much smaller than sex or class.

3f. Supporting variable: fare

```
In [19]: paid = df[df["Fare"] > 0]

fig, ax = plt.subplots(figsize=(8, 4.5))
sns.boxplot(data=paid, x="Class", y="Fare", hue="Outcome", order=CLASS_ORDER,
            hue_order=["Died", "Survived"], palette=OUTCOME_COLORS,
            fliersize=2, linewidth=1, saturation=1, ax=ax)
ax.set_yscale("log")
ax.yaxis.set_major_formatter(mtick.StrMethodFormatter("${x:,.0f}"))
ax.set_title("Survivors paid more, even within the same class")
ax.set_xlabel("Ticket class")
ax.set_ylabel("Fare (log scale)")
ax.legend(title="")
fig.savefig(FIG_DIR / "05_fare.png")
plt.show()

df.pivot_table(index="Class", columns="Outcome", values="Fare",
               aggfunc="median").round(2)
```



Out[19]:

	Outcome	Died	Survived
Class			
1st		44.75	77.96
2nd		13.00	21.00
3rd		8.05	8.52

Class		
1st	44.75	77.96
2nd	13.00	21.00
3rd	8.05	8.52

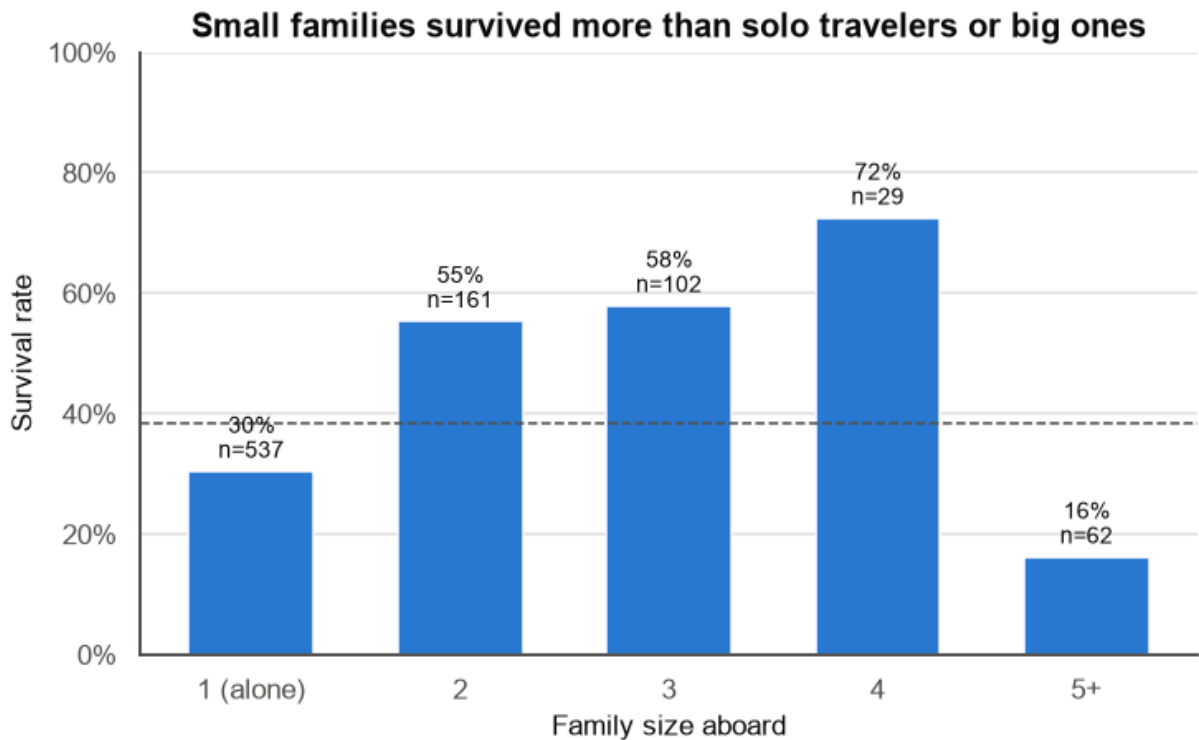
Observation: the median fare of survivors (26.00) is about 2.5 times that of passengers who died (10.50), but most of that is simply class. The box plot separates the two: inside first class, survivors still paid more (median 77.96 versus 44.75). The gap is smaller in second class (21.00 versus 13.00) and nearly disappears in third class (8.52 versus 8.05). Fare mostly acts as a stand-in for class and cabin location.

3g. Supporting variable: family size

```
In [20]: df["FamilyBand"] = pd.cut(df["FamilySize"], bins=[0, 1, 2, 3, 4, 20],
                                   labels=["1 (alone)", "2", "3", "4", "5+"])
fam = df.groupby("FamilyBand", observed=True)["Survived"].agg(
    ["mean", "size"])

fig, ax = plt.subplots(figsize=(7.5, 4.3))
ax.bar(fam.index.astype(str), fam["mean"], color=BLUE, width=0.6)
label_bars(ax, counts=fam["size"].tolist())
ax.axhline(BASE, ls="--", lw=1, color=MUTED)
pct_axis(ax, 1.0)
ax.set_title("Small families survived more than solo travelers or big ones")
ax.set_xlabel("Family size aboard")
ax.set_ylabel("Survival rate")
fig.savefig(FIG_DIR / "06_family.png")
plt.show()

fam.round(3)
```



Out[20]:

	mean	size
FamilyBand		
1 (alone)	0.304	537
2	0.553	161
3	0.578	102
4	0.724	29
5+	0.161	62

Observation: survival follows an upside-down U. Passengers traveling alone (30.4%, the largest group at 537) and families of five or more (16.1%) did worst, while families of two to four did best (55% to 72%). A likely reason is that many solo travelers were third-class men, and large families were mostly third class and harder to keep together during evacuation.

3h. Supporting variable: port of embarkation

```
In [21]: port_rate = df.groupby("Port")["Survived"].agg(["mean", "size"])
port_rate = port_rate.reindex(PORT_ORDER)
mix = pd.crosstab(df["Port"], df["Class"], normalize="index")
mix = mix.reindex(index=PORT_ORDER, columns=CLASS_ORDER)

fig, axes = plt.subplots(1, 2, figsize=(12, 4.5))

axes[0].bar(port_rate.index, port_rate["mean"], color=BLUE, width=0.55)
label_bars(axes[0], counts=port_rate["size"].tolist())
axes[0].axhline(BASE, ls="--", lw=1, color=MUTED)
pct_axis(axes[0], 1.0)
axes[0].set_title("Survival rate by port")
axes[0].set_ylabel("Survival rate")

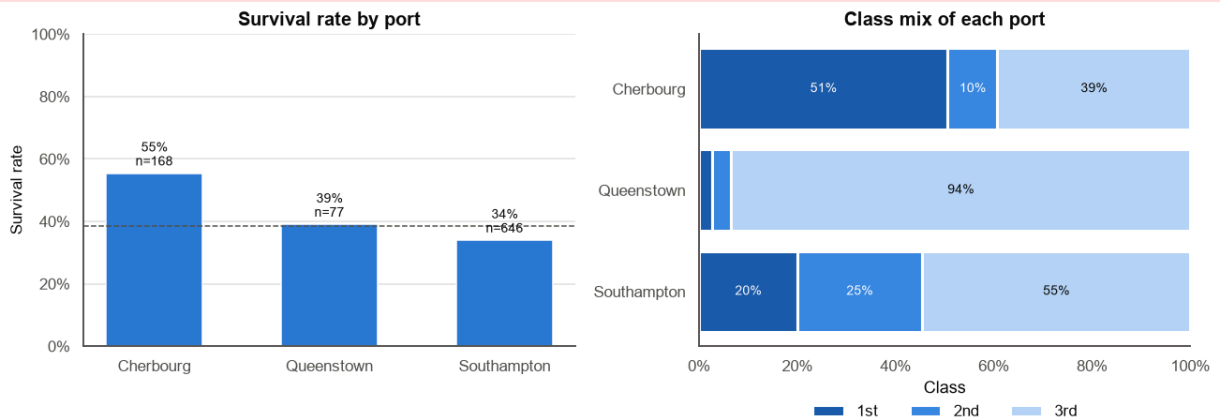
left = np.zeros(len(mix))
shades = {"1st": "#1c5cab", "2nd": "#3987e5", "3rd": "#b7d3f6"}
for cls in CLASS_ORDER:
    axes[1].barh(mix.index, mix[cls], left=left, color=shades[cls],
                 edgecolor="white", linewidth=2, label=cls)
    for y, (l, w) in enumerate(zip(left, mix[cls])):
        if w > 0.06:
            axes[1].text(l + w / 2, y, f"{w:.0%}", ha="center", va="center",
                        fontsize=9, color="white" if cls != "3rd" else INK)
    left += mix[cls].to_numpy()
axes[1].xaxis.set_major_formatter(mtick.PercentFormatter(1.0))
axes[1].set_xlim(0, 1)
axes[1].invert_yaxis()
axes[1].set_title("Class mix of each port")
axes[1].legend(title="Class", ncol=3, loc="lower center",
              bbox_to_anchor=(0.5, -0.28), frameon=False)
fig.tight_layout()
fig.savefig(FIG_DIR / "07_port.png")
```

```
plt.show()

port_rate.round(3)
```

C:\Users\Travis\AppData\Local\Temp\ipykernel_44112\3459431959.py:1: FutureWarning: The default of observed=False is deprecated and will be changed to True in a future version of pandas. Pass observed=False to retain current behavior or observed=True to adopt the future default and silence this warning.

```
port_rate = df.groupby("Port")["Survived"].agg(["mean", "size"])
```



Out[21]:

	mean	size
Port		
Cherbourg	0.554	168
Queenstown	0.390	77
Southampton	0.339	646

Observation: Cherbourg passengers survived at 55.4%, compared with 39.0% for Queenstown and 33.9% for Southampton. The class mix explains most of this. About half of Cherbourg's passengers (51%) were first class, compared with 20% at Southampton. Queenstown is the interesting exception: it was 94% third class yet sat near the overall rate, and it also had the highest share of women (47% versus 32% at Southampton), so sex is likely lifting it. Port looks like a signal but is mostly a proxy for class, which is a good reminder that a chart of one variable can mislead when another variable is doing the work.

3i. Correlation overview

```
In [22]: num = df.assign(Female=(df["Sex"] == "female").astype(int))
          ["Survived", "Female", "Pclass", "Age", "FamilySize", "IsAlone",
           "HasCabin", "FareLog"]
          corr = num.corr().round(2)

          mask = np.triu(np.ones_like(corr, dtype=bool), k=0)
          plot_corr = corr.iloc[1:, :-1]
          plot_mask = mask[1:, :-1]

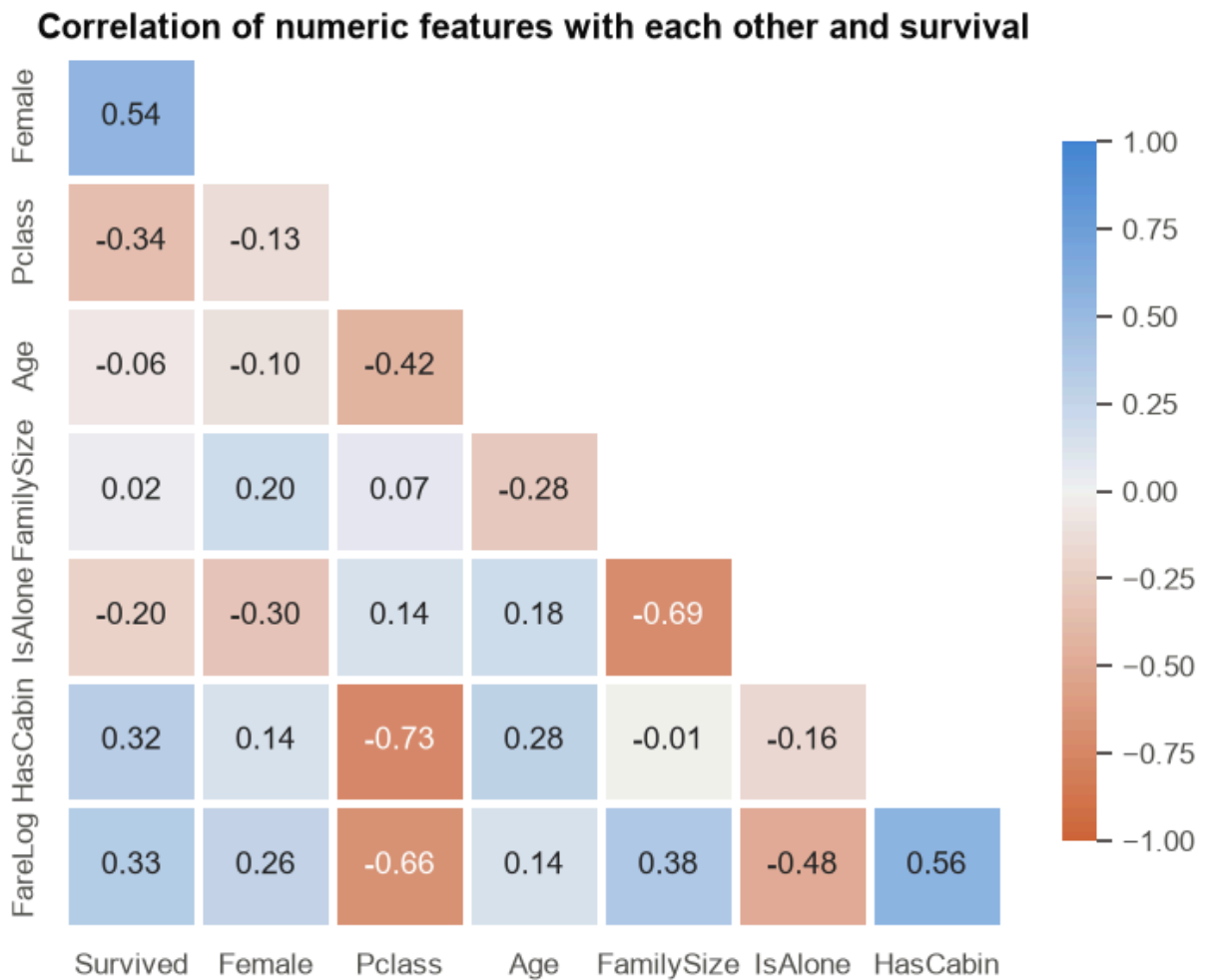
          fig, ax = plt.subplots(figsize=(8, 6))
```

```

cmap = sns.diverging_palette(25, 250, s=80, l=55, as_cmap=True)
sns.heatmap(plot_corr, mask=plot_mask, annot=True, fmt=".2f", cmap=cmap,
            center=0,
            vmin=-1, vmax=1, linewidths=2, linecolor="white",
            cbar_kws={"shrink": 0.8}, ax=ax)
ax.grid(False)
ax.set_title("Correlation of numeric features with each other and survival")
fig.savefig(FIG_DIR / "08_corr.png")
plt.show()

corr["Survived"].drop("Survived").sort_values(key=abs, ascending=False)

```



```

Out[22]: Female      0.54
Pclass    -0.34
FareLog    0.33
HasCabin   0.32
IsAlone   -0.20
Age        -0.06
FamilySize 0.02
Name: Survived, dtype: float64

```

Observation: the strongest correlation with survival is being female (+0.54), followed by ticket class (-0.34, since a higher class number means a lower class), log fare (+0.33), and having a recorded cabin (+0.32). Class, cabin, and fare are strongly correlated with each other (class with cabin at -0.73 and with log fare at -0.66), which confirms they mostly

measure the same thing: wealth and location on the ship. Age and family size have weak linear correlations because their effects are not straight lines, as the charts in 3e and 3g showed.

Step 4: Report

Results against the hypothesis

```
In [23]: f_rate = sex_rate.loc["Female", "mean"]
m_rate = sex_rate.loc["Male", "mean"]
report = pd.DataFrame([
    {"ID": "H1", "Claim": "Women survived more than men",
     "Evidence": f"Female {f_rate:.1%} vs male {m_rate:.1%}; V = 0.54",
     "Result": "Supported"},
    {"ID": "H2", "Claim": "Survival falls from 1st to 3rd class",
     "Evidence": " / ".join(f"{c} {r:.1%}" for c, r in
                           class_rate["mean"].items()) + "; V = 0.34",
     "Result": "Supported"},
    {"ID": "H3", "Claim": "Sex gap holds inside every class",
     "Evidence": "Women ahead in all 3 classes; 3rd-class women (50%) "
                 "beat 1st-class men (37%)",
     "Result": "Supported"},
])

for _, row in report.iterrows():
    print(f"{row['ID']} {row['Result'].upper(): {row['Claim']}")
    print(f"    Evidence: {row['Evidence']}")
    print()
```

```
H1 SUPPORTED: Women survived more than men
Evidence: Female 74.2% vs male 18.9%; V = 0.54

H2 SUPPORTED: Survival falls from 1st to 3rd class
Evidence: 1st 63.0% / 2nd 47.3% / 3rd 24.2%; V = 0.34

H3 SUPPORTED: Sex gap holds inside every class
Evidence: Women ahead in all 3 classes; 3rd-class women (50%) beat 1st-class men
(37%)
```

Summary of findings

1. **Sex was the strongest factor.** Women survived at 74.2% versus 18.9% for men. Sex had the largest effect size of any variable tested.
2. **Class was the second strongest factor.** Survival dropped from 63.0% in first class to 24.2% in third class.

3. **The two factors combine.** First-class women nearly all survived (97%), while second- and third-class men almost all died (14% to 16%). Even the worst-off women out-survived the best-off men.
4. **Age mattered mainly for males.** Boys under 13 survived at about 56%, while adult men survived at about 16%.
5. **Some apparent effects are really class.** Higher fares and the Cherbourg port both show higher survival, but both mostly reflect a larger share of first-class passengers.
6. **Traveling alone or in a large family was risky.** Solo travelers (30%) and families of five or more (16%) did worse than small families (55% to 72%).

Conclusion

The data supports the hypothesis. Women and passengers in higher ticket classes survived at much higher rates than men and passengers in lower classes, and the chi-square tests show these gaps are far too large to be chance. The analysis adds an important detail the original hypothesis did not state: sex outweighs class, so the full ordering runs from upper-class women, through lower-class women and upper-class men, down to lower-class men.

Limitations

- These are associations, not proof of cause. The charts describe who survived but cannot show why.
- 20% of ages were imputed from group medians, so age results rely mainly on the 714 recorded ages.
- The analysis set is 891 of the roughly 2,200 people aboard, so the rates describe this sample rather than every passenger and crew member.
- Some groups are small (for example 26 seniors and 44 teens), so their rates are less reliable.

```
In [24]: # Save the analysis set.
df.to_csv("titanic_analysis.csv", index=False)
print("Saved titanic_analysis.csv with shape", df.shape)
print("Figures saved:")
for png in sorted(FIG_DIR.glob("*.png")):
    print(" ", png.name)
```

Saved titanic_analysis.csv with shape (891, 26)

Figures saved:

```
01_overall.png
02_sex_class.png
03_heatmap.png
04_age.png
05_fare.png
06_family.png
07_port.png
08_corr.png
```

In []: